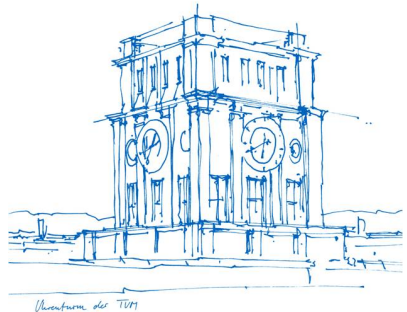


PoreAccess-CO₂: From Random Packing to an Interpretable Transport Closure

Structure-resolved diffusion and adsorption in
DEM-generated zeolite beds

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CO₂ adsorption matters and packing structure is a missing scale

Why CO₂ capturing matters?

- Food systems contribute about **one-third** of global anthropogenic greenhouse-gas emissions.
- Food and bioprocessing include recoverable biogenic CO₂ streams, for example from fermentation.
- Regenerable sorbents offer a compact capture and purification route.

What is already established?

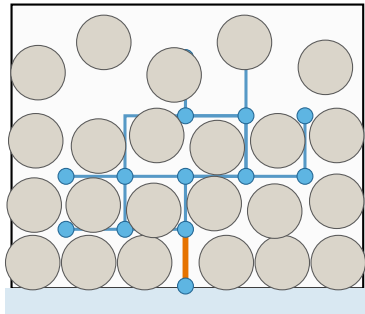
- CO₂ equilibrium and uptake kinetics on zeolite 13X are widely studied.
- Fixed-bed models normally use bulk porosity and a fitted effective transport coefficient.
- Random inlet-connected paths remain hidden inside that coefficient.

Novelty: connect random particle geometry to a fast bed-scale uptake model through a physically calculated pore-accessibility descriptor.

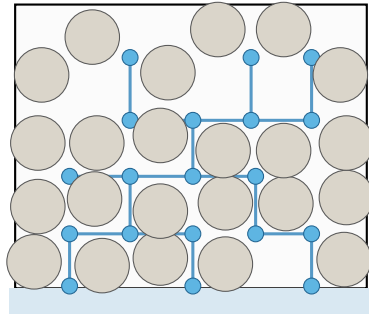
Same porosity can hide different inlet-connected pathways

Can a pore-accessibility measure predict bed-scale transient transport better than porosity alone?

Packing A: inlet bottleneck



Packing B: multiple inlet paths



- pore body
- diffusive throat
- zeolite bead
- inlet bottleneck

Same beads and similar porosity; different numbers and conductances of inlet-connected paths.

Why zeolite 13X and what does the study isolate?

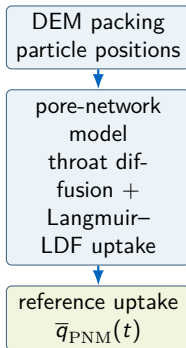
Why zeolite 13X?

- established benchmark sorbent for CO₂
- strong equilibrium uptake at low partial pressure
- commercial millimetre-scale beads form random beds

Controlled model problem

- dry, isothermal, stagnant gas
- CO₂ enters from a reservoir below the bed
- pore-network model resolves throat diffusion and bead adsorption
- no imposed flow; no CFD

How strongly does random void connectivity change access to the adsorbent?



Twenty beds isolate variability caused by random arrangement

Quantity	Fixed value
beads per bed	1850
bead diameter	1.6 mm
tube diameter	16 mm
D_{tube}/d_p	10
temperature	298.15 K
inlet c_{CO_2}	6.05 mol m^{-3}
simulation time	5000 s

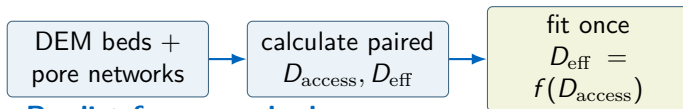
Only the DEM insertion seed changed

- Delaunay–Voronoi networks were built directly from particle coordinates
- all networks were connected and spanning
- quasi-uniform Sobol sampling allocated pore volumes without a voxel grid
- transient and 1D models passed mass-balance checks

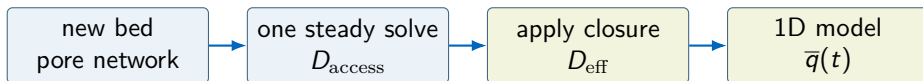
The ensemble tests arrangement effects under otherwise identical conditions.

Development and later use are two different workflows

Develop closure



Predict for a new bed



For a new bed, the transient pore-network adsorption model is no longer required.

Pore-network geometry and throat diffusion

Network construction

- four bead centres define a Delaunay tetrahedron; its circumcentre is a pore body
- tetrahedra sharing a triangular face are joined by a throat
- topology comes from particle coordinates, not voxels

A low-discrepancy Sobol sample of 2^{22} points allocates the analytically calculated void volume to pore bodies; it does not define connectivity.

Diffusion through throat $i-j$

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$$

$L_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$ is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area

$$A_{ij} = \pi r_{ij}^2,$$

where r_{ij} is the clearance of the shared Delaunay face.

The circular throat area is a geometric modelling assumption; G_{ij} has units of $\text{m}^3 \text{s}^{-1}$.

Langmuir equilibrium and bead uptake

Local equilibrium loading

$$p_k = c_k RT, \quad q_k^* = \frac{q_s b p_k}{1 + b p_k}.$$

Finite-rate uptake

$$\frac{dq_k}{dt} = k_{\text{LDF}} (q_k^* - q_k).$$

q_s is the saturation loading, b is the Langmuir affinity, p_k is the local CO_2 partial pressure, and q_k^* is the corresponding equilibrium loading. The product $b p_k$ is dimensionless.

The coupled diffusion–adsorption calculation conserves the total amount of CO_2 .

Meaning of the rate equation

- q_k is the current loading of bead k
- k_{LDF} controls how quickly q_k approaches q_k^*
- gas removed from adjacent pores is added to the bead

Pore-network outputs

- pore concentrations $c_i(t)$
- particle loadings $q_k(t)$
- mean uptake $\bar{q}_{\text{PNM}}(t)$

The homogeneous 1D uptake model

Axial diffusion and adsorption

$$\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t},$$

$$\frac{\partial q}{\partial t} = k_{\text{LDF}} [q^*(c) - q].$$

The bed is divided into 100 axial cells. A trial diffusivity D gives concentration $c(z, t)$ and loading $q(z, t)$.

Quantity compared with the network

$$\bar{q}_{\text{1D}}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz.$$

The 1D model uses the same bed height, porosity, inlet condition, Langmuir parameters, LDF coefficient and initial state as the pore network. Only D changes during fitting.

For every trial D , the 1D solver produces a complete mean uptake curve.

D_{access} and D_{eff} answer different questions

D_{access} : geometric accessibility

- fix concentration at inlet and interior plane
- solve one steady, linear diffusion problem
- no adsorption and no time integration

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

G_{access} is total steady flux per concentration difference. H is bed height; A_{tube} is tube area.

D_{eff} : transient reduced-model coefficient

- full transient network gives $\bar{q}_{\text{PNM}}(t)$
- repeated 1D solves give $\bar{q}_{1\text{D}}(t; D)$

Define the curve residual

$$e_m(D) = \bar{q}_{1\text{D}}(t_m; D) - \bar{q}_{\text{PNM}}(t_m).$$

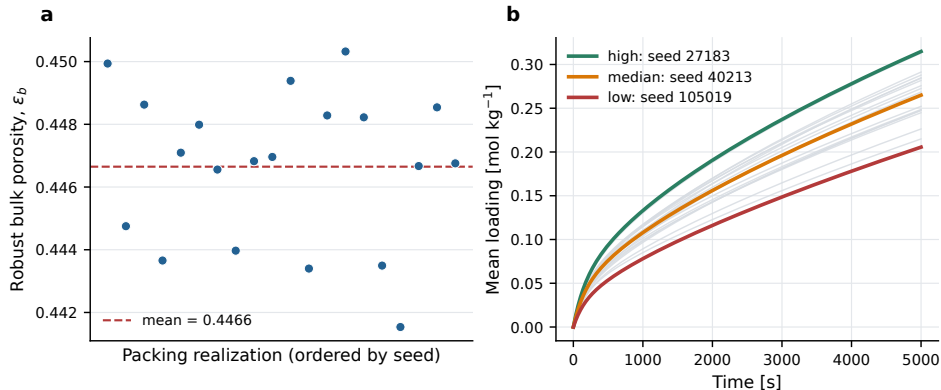
$$\text{NRMSE}(D) = \frac{\sqrt{\sum_m w_m e_m(D)^2}}{\bar{q}_{\text{PNM}}(t_f)},$$

$$D_{\text{eff}} = \arg \min_D \text{NRMSE}(D).$$

The trapezoidal weights w_m account for unequal time spacing and satisfy $\sum_m w_m = 1$.

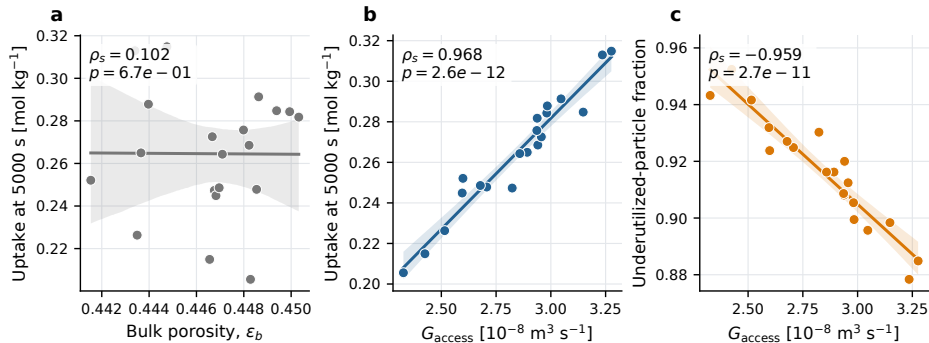
D_{access} is cheap because it avoids adsorption dynamics and optimization.

Nearly equal porosity produced clearly different uptake curves



Porosity CV: 0.55%. Final-uptake CV: 11.0% (range: 0.206–0.315 mol kg⁻¹).

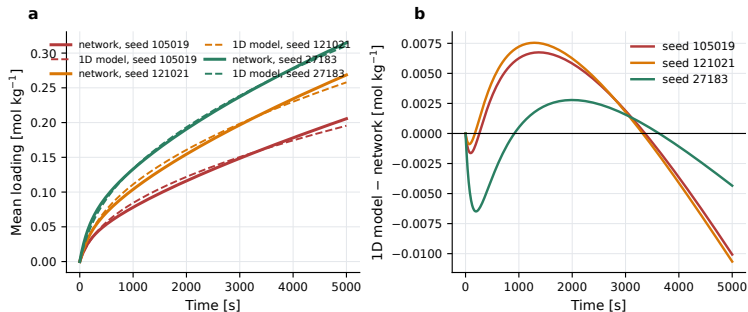
Accessibility explains the response that porosity misses



$$f_{\text{under}} = \frac{1}{N_p} \sum_{k=1}^{N_p} \mathbb{I}[q_k(5000 \text{ s}) < 0.5 q^*(p_{\text{inlet}})].$$

Higher G_{access} gives greater uptake and fewer particles below 50% of the inlet-equilibrium loading.

For each bed, D_{eff} is fitted to the complete uptake curve



(a) Solid lines: pore-network reference $\bar{q}_{\text{PNM}}(t)$ Dashed lines: 1D model $\bar{q}_{1\text{D}}(t; D_{\text{eff}})$

(b) Residual $\bar{q}_{1\text{D}} - \bar{q}_{\text{PNM}}$: positive values are overpredictions and negative values are underpredictions.

Across all 20 beds, the time-weighted NRMSE is 0.84–2.46%, with a mean of 1.58%.

The closure maps a cheap descriptor to the fitted coefficient

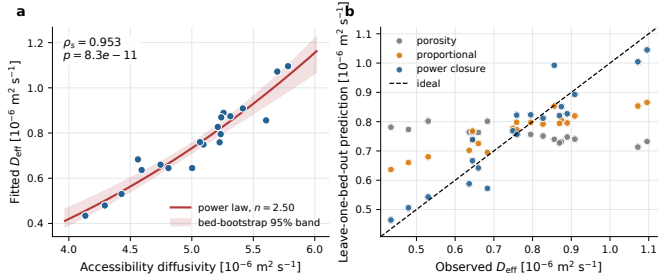
Closures compared

$$D_{\text{eff}} = a_0 + a_1 \varepsilon b,$$

$$D_{\text{eff}} = \alpha D_{\text{access}},$$

$$\frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n.$$

Complete beds are left out during fitting and then predicted.



(a) Selected power-law fit. ρ_s is the Spearman rank correlation; p tests the absence of a monotonic association.

(b) Held-out predictions from all three models; the proportional model is the orange series.

The positive power law gives LOOCV $R^2 = 0.890$; porosity gives $R^2 = -0.234$.

Why not predict uptake directly from G_{access} ?

Direct endpoint relation

$$G_{\text{access}} \longrightarrow \bar{q}(5000 \text{ s})$$

- strong correlation: $\rho_s = 0.968$
- ranks beds or predicts one endpoint
- valid at the selected time and conditions

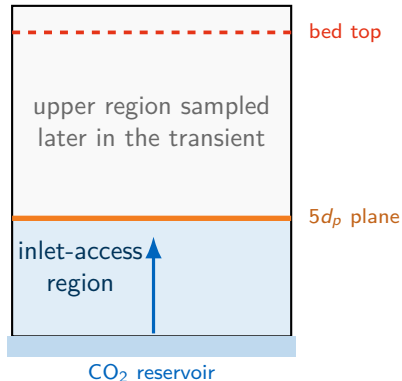
Transport closure

$$G_{\text{access}} \longrightarrow D_{\text{access}} \longrightarrow D_{\text{eff}} \longrightarrow \bar{q}(t)$$

- generates the complete uptake curve
- retains conservative 1D transport
- gives axial $c(z, t)$ and $q(z, t)$ profiles

If only uptake at 5000 s is needed, G_{access} is sufficient; D_{eff} is needed for a reusable transient model.

Why use an interior plane?



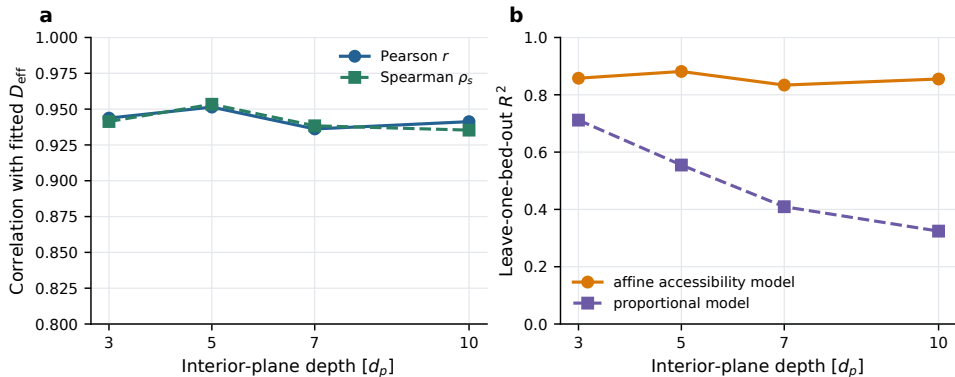
A plane at $5d_p$ includes

- the inlet openings
- second-layer and deeper bottlenecks
- parallel paths connecting inlet to the plane

The bed top answers a different question

- measure conductance through the entire bed
- include upper bottlenecks not sampled early

The conclusion is stable from $3d_p$ to $10d_p$



$5d_p$ is retained because it lies beyond the immediate inlet layer and gives the strongest prediction.

What the present evidence establishes

Supported

- porosity alone misses important packing variability
- inlet-connected topology controls transient utilization
- one D_{eff} reproduces each detailed curve accurately
- D_{access} provides an interpretable predictive closure

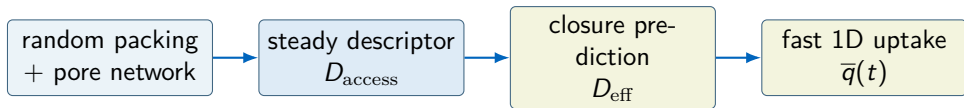
Not yet claimed

- industrial breakthrough performance
- effects of flow, heat or water competition
- transfer to other bead sizes or D_{tube}/d_p
- a universal power-law exponent

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

Pore accessibility retains what bulk porosity loses

Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



20 beds
controlled ensemble

1.58%
mean reduction error

$R^2 = 0.890$
withheld-bed closure